

Install Intel® VTune™ Profiler

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Intel® VTune™ Profiler Installation Guide

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Before You Begin

The information in this installation guide is applicable to the latest version of VTune Profiler.

Depending on your operating environment, available hardware, and typical workflow, you can install VTune Profiler in several ways, which this topic describes.

- **Basic Terminology:**

- **Host :** This is the system where you install the VTune Profiler application. Typically, this is also the system where you view the results of a data collection.
- **Target :** This is the system where you run data collection on a software application. If the host and target systems are different, you copy results to the host after data collection to view them.
- **Local collection :** You install VTune Profiler on a system and run a data collection on an application that is on the same system. The host and target locations are the same.
- **Remote collection :** You install VTune Profiler on a system and run a data collection on an application that is on a different system. The host and target locations are different. The download package for VTune Profiler contains zip files of collectors for target packages. Before you run a remote data collection, you install a collector on the target system to enable data collection. Once the data collection completes, VTune Profiler transfers results to the host system for viewing.

- **Operating System (Host):**

You can install VTune Profiler on [Windows](#) and [Linux](#) host systems.

- **Operating System (Target):**

You can profile target applications on these systems:

- [Windows](#)
- [Linux](#)
- [FreeBSD*](#)
- [Android*](#)
- [QNX*](#)

- **Installation Package:**

You install VTune Profiler as a single package to use with the GUI and command line interface (CLI). Install VTune Profiler through one of these ways:

- [As a standalone package](#)
- With the [Intel® oneAPI Base Toolkit](#)

- **Installation Workflow:**

1. Download the latest version of VTune Profiler as a single package (standalone version or with the Intel® oneAPI Base toolkit).
2. Install VTune Profiler on the host system.
3. For remote collections, copy the relevant target collector (zip file) to the target system. You do not need to install the full package of VTune Profiler on the target side.
4. Before you run a remote collection, extract the collector on the target system. With some targets, VTune Profiler can deploy the collector for you on the target. During remote collection, VTune Profiler locates the collector on the target system and uses it for data collection.

- **Types of Installation:**

Use this table to see available installation options for your target operating system.

NOTE Before you run a remote data collection, make sure to [install collectors for the remote targets](#) from the VTune Profiler download package.

		Type of Collection	Profile Applications on This Target				
			Windows	Linux	FreeBSD	Android	QNX
Install VTune Profiler on This Host	Windows	Local	GUI CLI	N/A	CLI*	N/A	N/A
		Remote	Use Browser	GUI CLI	GUI CLI	GUI CLI	CLI
	Linux	Local	N/A	GUI CLI	CLI*	N/A	N/A
		Remote	Use Browser (View Results Only)	GUI CLI	GUI CLI	GUI CLI	CLI

* : Once you unzip and run the FreeBSD collector on the remote target, you can run a data collection from the command line within the target. Transfer the results to your host system to view them.

- **Install a Specific Version of VTune Profiler**

For maximum benefit, download (or upgrade to) the latest version of VTune Profiler. If you have purchased a license of VTune Profiler with [priority support](#), contact your Intel representative to get access to older versions of VTune Profiler.

- **Operating Models:**

Use VTune Profiler in any of these operating models:

- Install a standalone package ([Windows](#) or [Linux](#) systems)
- Profile through target packages
- Use VTune Profiler as a Server
- In a cloud environment
- In containers
- On a virtual machine
- Integrate into continuous integration (CI) pipeline

To learn more, see [Operating Models](#).

Install on Windows* OS

Install Intel® VTune™ Profiler on a Windows system, either as a standalone application or as a component in the Intel® oneAPI Base Toolkit.

Before you Begin

Check Requirements:

- Your host system should use a 64-bit OS.
- Ensure that you are using supported versions of Windows and compilers. Check [System Requirements](#).

Get Administrative Privileges

Administrative privileges are necessary for proper installation of the [Sampling Drivers](#) that are required for many analysis types in VTune Profiler. Having administrative privileges also makes it easier to install, change, or uninstall VTune Profiler.

Download VTune Profiler

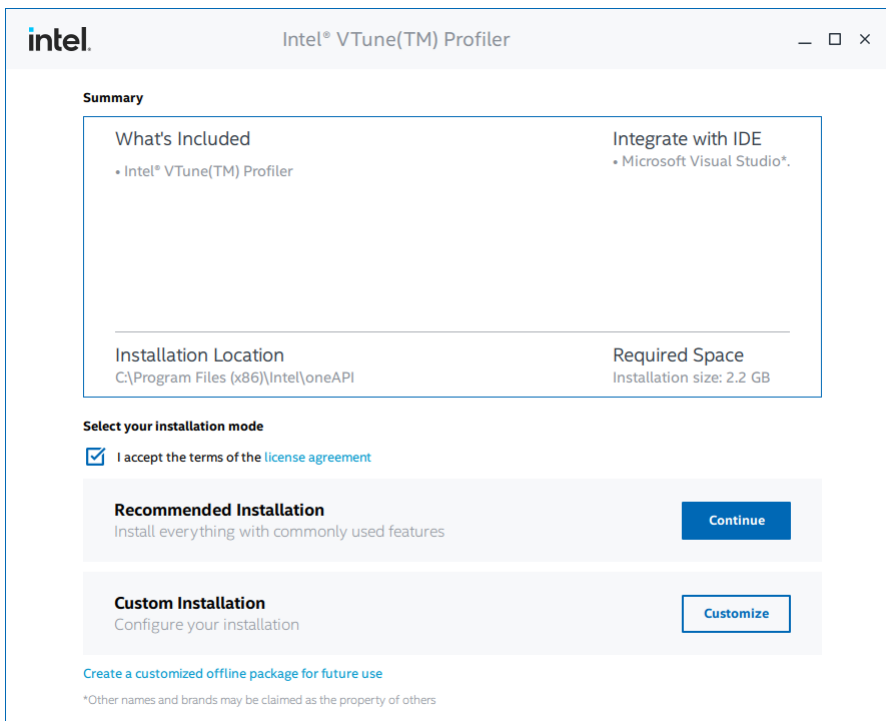
Download Intel® VTune™ Profiler for Windows OS as:

- [A standalone application](#)
- [A component in the Intel® oneAPI Base Toolkit](#)

Install VTune Profiler as a Standalone Application

On the [download page for VTune Profiler](#),

1. Select between an Online or an Offline installer. An offline installer is better suited for systems without a stable internet connection.
2. Once the download completes, run the installer executable with Administrative privileges.
3. Select your preferred mode for the installation.

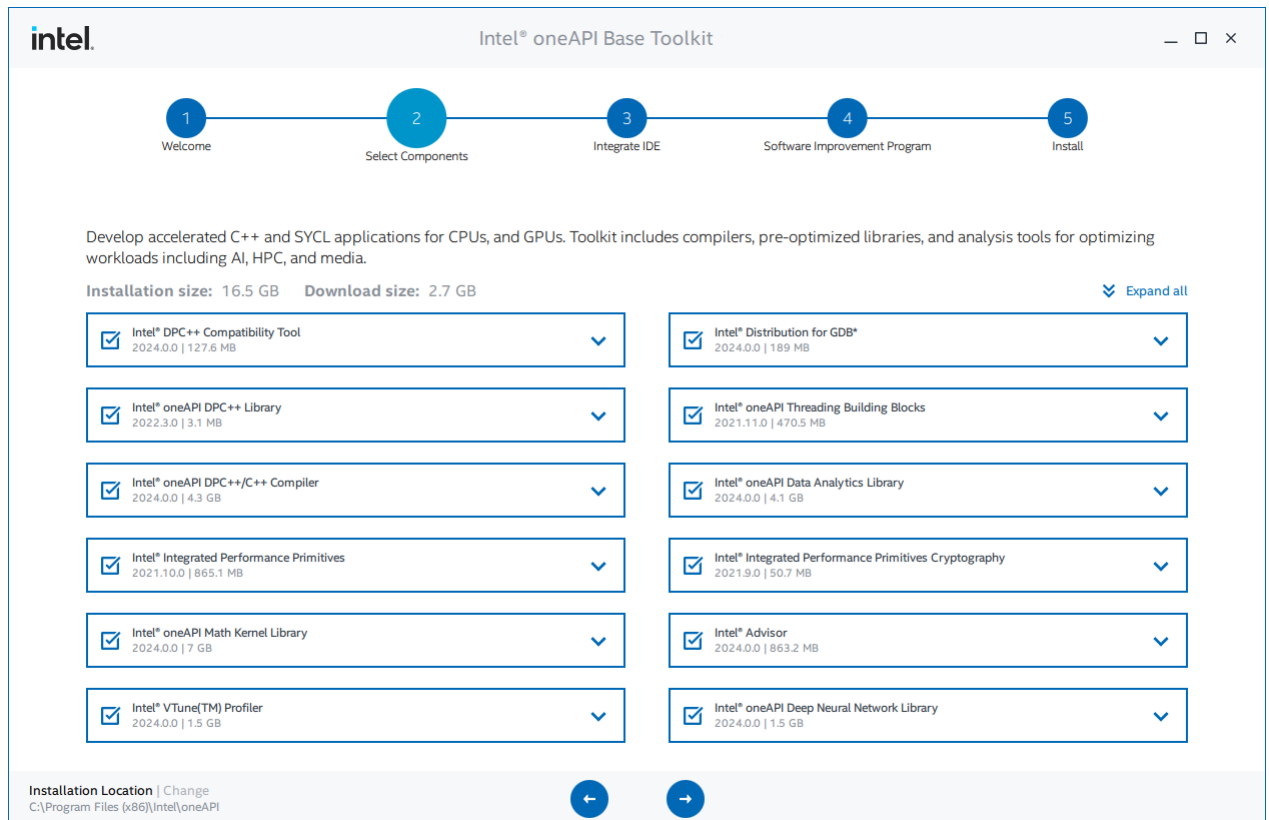


- In the **Recommended Installation** mode, VTune Profiler installs in the default directory and integrates into your Microsoft Visual Studio* IDE.
 - In the **Custom Installation** mode, you can control these settings:
 - **Install directory:** Specify a custom install directory. The default location is <Program Files (x86)>\Intel\oneAPI\vtune\
 - **Microsoft Visual Studio* IDE integration:** Choose a version of Microsoft Visual Studio for integration, if necessary.
4. Follow subsequent steps in the installer to complete the installation.

Install VTune Profiler as a Component in the Intel® oneAPI Base Toolkit

On the [download page for Intel® oneAPI Base Toolkit](#):

1. Select between an Online or an Offline installer. An offline installer is better suited for systems without a stable internet connection.
2. Once the download completes, run the installer executable with Administrative privileges.
3. Select your preferred mode for the installation.
 - Choosing the **Recommended Installation** mode installs all components of the Intel® oneAPI Base Toolkit in the default directory. Appropriate components are then integrated into your Microsoft Visual Studio* IDE.
 - In the **Custom Installation** mode, you can control these settings:
 - **Components to install:** Choose the toolkit components you want to install. The installer first checks for components that are already installed on your system (if any). You have the option to update or skip those components.
 - **Install directory:** Specify a custom install directory. The default location is <Program Files (x86)>\Intel\oneAPI\
 - **Microsoft Visual Studio* IDE integration:** Choose a version of Microsoft Visual Studio for integration, if necessary.



4. Follow subsequent steps in the installer to complete the installation.

After you Install VTune Profiler

Set Environment Variables:

To use the command line interface of VTune Profiler, set environment variables first. Run the `vars.bat` script:

```
C:"Program Files (x86)"\Intel\oneAPI\vtune\latest\env\vars.bat
```

Verify the Sampling Drivers:

The installation of VTune Profiler on your Windows system should also install the sampling drivers necessary for data collection in most analysis types. However, it is always good practice to verify this:

1. Open the command prompt as an administrator.
2. Go to the `bin64` folder in your installation directory.
3. Check to make sure your system meets the requirements for hardware event-based sampling (HEBS) collection. Run this command:

```
amplxe-sepreg.exe -c
```

4. Check if the sampling drivers are loaded. Run:

```
amplxe-sepreg.exe -s
```

5. If your system meets the requirements for HEBS collection and the sampling drivers are not installed, run this command to install the drivers with administrative privileges:

```
amplxe-sepreg.exe -i
```

For more information on installing the sampling drivers on a Windows system, see [here](#).

Verify Your Installation:

To verify if you installed VTune Profiler correctly or troubleshoot issues with the installation, use the self-check script (`vtune-self-checker.bat`) available in the package. This script runs a set of analyses on a sample application, and reports the progress interactively.

1. Set environment variables.
2. [Verify/Install the sampling drivers](#).
3. Open a terminal window.
4. Go to this location: `<Program Files (x86)>\Intel\oneAPI\vtune\latest\bin64`
5. Run the `vtune-self-checker.bat` script:

```
vtune-self-checker.bat
```

The self-check script may take several minutes to run all necessary checks.

6. Once the script completes all checks, review the summary. Check if any analyses failed, and see the log file for additional information.

Prepare Remote Targets (Optional):

When you install VTune Profiler on a Windows platform, you can use it to profile and collect data remotely from non-Windows targets, such as Linux* machines. For more information, see [Set Up Analysis Target](#).

Learn More

- See guided examples in the [Get Started Guide](#).
- Follow a comprehensive tutorial to [Analyze Common Performance Bottlenecks](#)
- Learn about analysis types, workflows, and command line use in the [User Guide](#)
- Browse through technical recipes that address typical use cases and challenges in the [Performance Analysis Cookbook](#)

Install on Linux* OS

Install Intel® VTune™ Profiler on a Linux machine through these ways.*

Choose how you want to install Intel® VTune™ Profiler:

- [Use a Regular Installer \(Online or Offline\) for the Standalone Package or Intel® oneAPI Toolkit](#)
- [Use Package Managers \(APT, YUM, DNF, Zypper\)](#)

NOTE

Occasionally, updates to supported operating systems can cause the [sampling drivers](#) to fail during building or loading. In this case, before you update Intel® VTune™ Profiler, you may need to [download the latest versions](#) of the sampling drivers and manually [build the drivers](#).

Install with a Regular Installer

Install Intel® VTune™ Profiler on a Linux system, either as a standalone application or as a component in the Intel® oneAPI Base Toolkit. Use the online or offline installers.

Before you Begin

Check Requirements:

- Your host machine should use a 64-bit operating system. Once you install VTune Profiler, you can use it to profile 32-bit and 64-bit applications.
- Ensure that you are using supported versions of Linux distributions and compilers. Check [System Requirements](#).
- Ensure that you have installed build tools (`make` and `gcc`) on your system. These are necessary to build and load the VTune Profiler drivers.
- Have the right permissions to use VTune Profiler drivers. To ensure this, add your account to the `vtune` group. Run:

```
$ sudo usermod -aG vtune <username>
```

Then log out and log back in for the addition to happen.

Get Superuser Privileges

Superuser privileges are necessary for proper installation of the [Sampling Drivers](#) that are required for many analysis types in VTune Profiler. Having superuser privileges also makes it easier to install, change, or uninstall VTune Profiler.

However, you can install VTune Profiler as a regular user and use most analysis types through the capabilities of Linux Perf*. For more information, see this technical recipe on [Profiling Hardware Without Intel Sampling Drivers](#).

Download VTune Profiler

Download Intel® VTune™ Profiler for Linux OS as:

- [A standalone application](#)
- [A component in the Intel® oneAPI Base Toolkit](#)

Install VTune Profiler as a Standalone Application

On the [download page for VTune Profiler](#),

1. Select between an Online or an Offline installer. An [offline installer](#) is better suited for systems without a stable internet connection.
2. Once the download completes, open a terminal window.
3. Go to the directory where you downloaded the installer.
4. If necessary, enable the execution of the installer. Run this command:

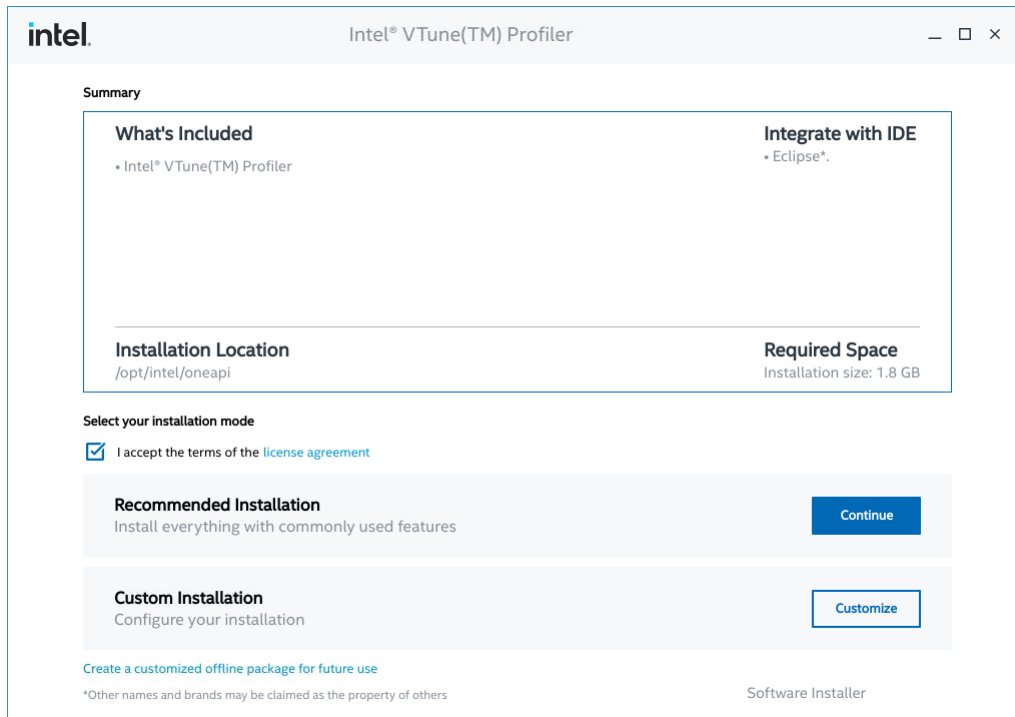
```
chmod +x <installer-package-name>.sh
```


5. Run the installer executable as a superuser:

```
sudo ./<installer-package-name>.sh
```

If a Linux desktop is available, the installer opens as a GUI. Otherwise, the installer continues in the terminal.

6. Select your preferred mode for the installation.



- In the **Recommended Installation** mode, VTune Profiler installs in the default directory and integrates into your Eclipse* IDE.
- In the **Custom Installation** mode, you can control these settings:
 - **Install directory:** Specify a custom install directory. The default location is `/opt/intel/oneapi/vtune/`
 - **Eclipse* IDE integration:** Choose a version of Eclipse for integration, if necessary.

7. Follow subsequent steps in the installer to complete the installation.

Install VTune Profiler as a Component in the Intel® oneAPI Base Toolkit

On the [download page for Intel® oneAPI Base Toolkit](#):

1. Select between an Online or an Offline installer. An [offline installer](#) is better suited for systems without a stable internet connection.
2. If necessary, enable the execution of the installer. Run this command:

```
chmod +x <installer-package-name>.sh
```

3. Run the installer executable as a superuser:

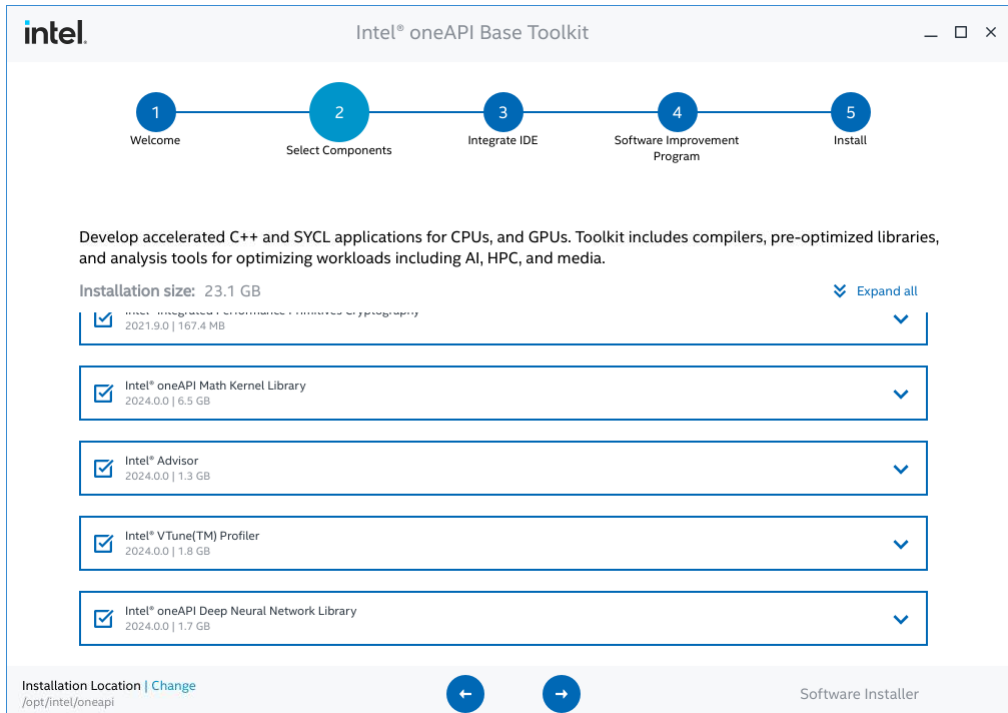
```
sudo ./<installer-package-name>.sh
```

If a Linux desktop is available, the installer opens as a GUI. Otherwise, the installer continues in the terminal.

4. Select your preferred mode for the installation.

- Choosing the **Recommended Installation** mode installs all components of the Intel® oneAPI Base Toolkit in the default directory. Appropriate components are then integrated into your Eclipse* IDE.

- In the **Custom Installation** mode, you can control these settings:
 - **Components to install:** Choose the toolkit components you want to install. The installer first checks for components that are already installed on your system (if any). You have the option to update or skip those components.
 - **Install directory:** Specify a custom install directory. The default location is `/opt/intel/oneapi/`
 - **Eclipse* IDE integration:** Choose a version of Eclipse for integration, if necessary.



5. Follow subsequent steps in the installer to complete the installation.

Use the Offline Installer for VTune Profiler

When your host system does not have a stable connection to the internet, use the offline installer to set up VTune Profiler.

On the download page for the [VTune Profiler standalone package](#) or the [Intel® oneAPI Base Toolkit](#),

1. Select the Offline Installer mode.
2. Download the offline installation package from the page. To download directly from the command line, copy the `wget` and `sudo` commands from the **Command Line Download** section at the bottom of the page.
3. Run the installer script.
4. When the installation process begins, you see options to:
 - **Accept & install**
 - **Accept & customize**
 - **Download only**
 - **Decline & quit**

If you want to change the default installation directory, choose to **Accept & customize** and set the installation directory.

5. The installer checks for available graphics packages in your Linux distribution. These packages are necessary to display the GUI in a Linux environment. A warning displays if the graphics packages are missing. If you plan to use VTune Profiler from the command line only, you can ignore this warning.

6. Choose if you want to integrate VTune Profiler within the Eclipse* IDE.
7. Complete the installation.

After you Install VTune Profiler

Set Environment Variables:

To use the command line interface of VTune Profiler, set environment variables first. Run these scripts:

bash:

```
source /opt/intel/oneapi/vtune/latest/env/vars.sh
```

csh/tcsh:

```
source /opt/intel/oneapi/vtune/latest/env/vars.csh
```

Verify the Sampling Driver:

The installation of VTune Profiler on your Linux system should also install the sampling drivers necessary for data collection in most analysis types. However, it is always good practice to verify this:

1. Open the command prompt as an administrator.
2. Go to the <install-dir>/sepdk/src folder in your installation directory.
3. Check if the sampling drivers are loaded. Run:

```
$ cd <install-dir>/sepdk/src
$ ./insmod-sep -q
```

If the drivers are loaded, the result of this command should inform you about group ownership and file permissions on the driver device.

4. If the drivers are unavailable, you must complete additional steps to [build and install the sampling drivers](#).

Verify Your Installation:

To verify if you installed VTune Profiler correctly or troubleshoot issues with the installation, use the self-check script (`vtune-self-checker.sh`) available in the package. This script runs a set of analyses on a sample application, and reports the progress interactively.

1. [Set environment variables](#).
2. [Verify/Install the sampling driver](#).
3. Open a terminal window.
4. Go to this location: `/opt/intel/oneapi/vtune/latest/bin64/`
5. Run the `vtune-self-checker.sh` script:

```
vtune-self-checker.sh
```

The self-check script may take several minutes to run all necessary checks.

6. Once the script completes all checks, review the summary. Check if any analyses failed, and see the log file for additional information.

Configure Data Collection Without Sampling Drivers (Optional):

If you install VTune Profiler without superuser privileges, you must complete additional configuration steps to enable hardware analyses through Linux Perf*.

To learn about these steps and possible limitations, see this technical recipe on [Profiling Hardware Without Intel Sampling Drivers](#).

Prepare Remote Targets (Optional):

When you install VTune Profiler on a Linux platform, you can use it to profile and collect data remotely from non-Linux targets. For more information, see [Set Up Analysis Target](#).

Learn More

- See guided examples in the [Get Started Guide](#).
- Follow a comprehensive tutorial to [Analyze Common Performance Bottlenecks](#)
- Learn about analysis types, workflows, and command line use in the [User Guide](#)
- Browse through technical recipes that address typical use cases and challenges in the [Performance Analysis Cookbook](#)

Install with Package Managers

On Linux OS, learn how to install Intel® VTune™ Profiler using the APT, YUM, DNF, or Zypper package managers.*

Before You Begin

- Make sure that your operating system is supported. Check [System Requirements](#).
- If you have not installed Intel® oneAPI software products using package managers previously, configure your package manager to work with the repository. Follow instructions in the sections below.

APT

Add the Intel oneAPI Repository

If you have not installed Intel® oneAPI Toolkits or components previously, configure your package managers to use Intel repositories:

1. (Optional) Switch to any directory with write access. This example uses /tmp:

```
cd /tmp
```

2. Download the Intel repository public key. Use wget:

```
wget https://apt.repos.intel.com/intel-gpg-keys/GPG-PUB-KEY-INTEL-SW-PRODUCTS.PUB
```

3. Add the public key to your apt keyring:

```
sudo apt-key add GPG-PUB-KEY-INTEL-SW-PRODUCTS.PUB
```

4. (Optional) Remove the public key file, since this is no longer required:

```
rm GPG-PUB-KEY-INTEL-SW-PRODUCTS.PUB
```

5. Configure the APT client to use the Intel repository:

```
echo "deb https://apt.repos.intel.com/oneapi all main" | sudo tee /etc/apt/sources.list.d/oneAPI.list
```

If the add-apt-repository utility is installed, run this command instead:

```
sudo add-apt-repository "deb https://apt.repos.intel.com/oneapi all main"
```

Install the VTune Profiler Package

1. Update the APT package lists:

```
sudo apt update
```

2. Install the VTune Profiler standalone package:

```
sudo apt install intel-oneapi-vtune
```

YUM

Add the Intel oneAPI Repository

If you have not installed Intel® oneAPI Toolkits or components previously, configure your package managers to use Intel repositories:

1. In a directory that has write access, create a YUM/DNF `.repo` file. This example uses `/tmp`:

```
tee > /tmp/oneAPI.repo << EOF
[oneAPI]
name=Intel® oneAPI repository
baseurl=https://yum.repos.intel.com/oneapi
enabled=1
gpgcheck=1
repo_gpgcheck=1
gpgkey=https://yum.repos.intel.com/intel-gpg-keys/GPG-PUB-KEY-INTEL-SW-PRODUCTS.PUB
EOF
```

2. Move the `oneAPI.repo` file that you just created, to the YUM/DNF configuration directory (`/etc/yum.repos.d`):

```
sudo mv /tmp/oneAPI.repo /etc/yum.repos.d
```

Install the VTune Profiler Package

Run this command:

```
sudo yum install intel-oneapi-vtune
```

DNF

Add the Intel oneAPI Repository

If you have not installed Intel® oneAPI Toolkits or components previously, configure your package managers to use Intel repositories:

1. In a directory that has write access, create a YUM/DNF `.repo` file. This example uses `/tmp`:

```
tee > /tmp/oneAPI.repo << EOF
[oneAPI]
name=Intel® oneAPI repository
baseurl=https://yum.repos.intel.com/oneapi
enabled=1
gpgcheck=1
repo_gpgcheck=1
gpgkey=https://yum.repos.intel.com/intel-gpg-keys/GPG-PUB-KEY-INTEL-SW-PRODUCTS.PUB
EOF
```

2. Move the `oneAPI.repo` file that you just created, to the YUM/DNF configuration directory (`/etc/yum.repos.d`):

```
sudo mv /tmp/oneAPI.repo /etc/yum.repos.d
```

Install the VTune Profiler Package

Run this command:

```
sudo dnf install intel-oneapi-vtune
```

Zypper

Add the Intel oneAPI Repository

If you have not installed Intel® oneAPI Toolkits or components previously, configure your package managers to use Intel repositories.

Run this command to add the Intel repository key:

```
sudo zypper addrepo https://yum.repos.intel.com/oneapi oneAPI
```

Install the VTune Profiler Package

Run this command:

```
sudo zypper install intel-oneapi-vtune
```

After you Install VTune Profiler

Set Environment Variables:

To use the command line interface of VTune Profiler, set environment variables first. Run these scripts:

bash:

```
source /opt/intel/oneapi/vtune/latest/env/vars.sh
```

csh/tcsh:

```
source /opt/intel/oneapi/vtune/latest/env/vars.csh
```

Verify Your Installation:

To verify if you installed VTune Profiler correctly or troubleshoot issues with the installation, use the self-check script (`vtune-self-checker.sh`) available in the package. This script runs a set of analyses on a sample application, and reports the progress interactively.

1. Before you run the self-check script, make sure to set environment variables.
2. Next, open a terminal window.
3. Go to this location: `/opt/intel/oneapi/vtune/latest/bin64/`
4. Run the `vtune-self-checker.sh` script:

```
vtune-self-checker.sh
```

The self-check script may take several minutes to run all necessary checks.

5. Once the script completes all checks, review the summary. Check if any analyses failed, and see the log file for additional information.

Configure Data Collection Without Sampling Drivers (Optional):

If you install VTune Profiler without superuser privileges, you must complete additional configuration steps to enable hardware analyses through Linux Perf*.

To learn about these steps and possible limitations, see this technical recipe on [Profiling Hardware Without Intel Sampling Drivers](#).

Learn More

- See guided examples in the [Get Started Guide](#).
- Follow a comprehensive tutorial to [Analyze Common Performance Bottlenecks](#)
- Learn about analysis types, workflows, and command line use in the [User Guide](#)
- Browse through technical recipes that address typical use cases and challenges in the [Performance Analysis Cookbook](#)

Operating Models

Learn how to use Intel® VTune™ Profiler in operating models that cater to various workflows.

- [Install VTune Profiler on Supported Platforms](#)
- [Use VTune Profiler as a Server \(Web Browser\)](#)
- [Use VTune Profiler in a Cloud Environment](#)
- [Use VTune Profiler in a Container](#)
- [Install VTune Profiler on a Virtual Machine](#)
- [Integrate VTune Profiler into your Continuous Integration \(CI\) Pipeline](#)

- [Profile with a Target Package](#)

Install VTune Profiler on Supported Platforms

Install VTune Profiler on these platforms:

- [Windows*](#)
- [Linux*](#)

You can install VTune Profiler :

- [As a standalone package](#)
- As a component in the [Intel® oneAPI Base Toolkit](#)

Learn More:

- [Release Notes](#) about features introduced in versions of VTune Profiler
- Technical compatibility information in [System Requirements](#)

Use VTune Profiler as a Server (Web Browser)

Purpose:

Run VTune Profiler as a web service. Access the application and run profiling analyses through a web browser.

Benefits:

- Access VTune Profiler in remote test environments.
- Run VTune Profiler on the target or remote development machine(s) without having to install the standalone package on any of them.
- Manage fewer system requirements in comparison to the standalone package.
- Share collection results through a URL.
- Deploy into your web hosting infrastructure or as a standalone service. VTune Profiler Server supports integration with identity providers through SAML SSO.
- Profile a pool of targets by sharing data collection agents.

Learn More:

- [Install VTune Profiler Server](#)

Use VTune Profiler In a Cloud Environment

Purpose:

Install VTune Profiler in public cloud instances from these Cloud Service Providers (CSPs):

- Amazon Web Service* (AWS)
- Google Cloud Platform*
- Microsoft Azure*

Benefits:

- Run user-mode analyses (Hotspots and Threading) in virtualized environments.
- Run hardware event-based analysis types on bare-metal instances.

Learn More:

- [Cloud Targets](#)
- A technical recipe on [Profiling Applications in Amazon Web Services* \(AWS\) EC2 Instances](#)
- [Intel® VTune™ Profiler Functionality on AWS* Instances](#)

Use VTune Profiler In a Container

Purpose:

Use VTune Profiler in popular containerized environments like Docker*.

Benefits:

- Run VTune Profiler inside the container.
- Analyze containerized workloads.
- Analyze a workload within the same container.
- Analyze non-containerized workloads while running inside container.
- Analyze multiple containers simultaneously.

Learn More:

- A technical recipe on [Profiling Docker* Containers](#)

Install VTune Profiler on a Virtual Machine

Purpose:

Install VTune Profiler on a virtual machine in the same way you would install on a bare-metal machine. Find support for these hypervisors:

- [VMware*](#)
- [Parallels*](#)
- [KVM*](#)
- [Xen*](#)
- [Hyper-V*](#)

To ensure maximum compatibility, you may need to complete additional configuration for each of these hypervisors.

Also, due to the inherent nature of virtualization, some limitations may exist in the way you access hardware performance counters.

Learn More:

- [Targets in Virtualized Environments](#)

Integrate VTune Profiler into your Continuous Integration (CI) Pipeline

Purpose:

Integrate VTune Profiler into your Continuous Integration pipeline to collect performance data in specific conditions.

Benefits:

- Track workload performance over time.
- Spot and investigate performance regressions.
- Use the command line interface (CLI) to run an analysis type, or a combination of analysis types, on your workload.

Learn More:

- A technical recipe about [Enabling Performance Profiling in GitLab* CI](#)

Profile With a Target Package

Purpose:

A target package is a minimal distribution of VTune Profiler. You use target packages to enable profiling capabilities on specific remote targets. With target packages, you do not install VTune Profiler on the target system. Instead, you copy the target package to the target system and apply additional configurations to enable remote profiling from a natively supported system.

In the case of FreeBSD* OS, you can also use VTune Profiler directly on the FreeBSD system through the [command line interface](#).

Benefits:

- Profile remote targets without installing the complete package of VTune Profiler.
- For FreeBSD targets, run command line analyses directly.

Learn More:

- [Install Collectors for Remote Targets](#)
- [FreeBSD* Targets](#)
- [QNX* Targets](#)
- [Android Targets](#)

Install Collectors for Remote Targets

Deploy data collectors to profile remote targets without installing the full package of Intel® VTune™ Profiler on the targets.

The downloadable package of VTune Profiler contains packages of data collectors for remote targets. When you want to perform a remote collection from a host machine, you deploy these collectors on the targets. You do not need to install the full package of VTune Profiler on the targets. You do need to have VTune Profiler installed on the host machine to view the collected results.

Depending on the target, you can also run a data collection directly on the target from the command line interface (CLI).

Install the Collectors

Windows* Installation:

Find the collector packages here:

```
C:\Program Files (x86)\Intel\oneAPI\vtune\latest\target
```

Linux* Installation:

Find the collector packages here:

```
/opt/Intel/oneAPI/vtune/latest/target
```

Copy the collector packages from these locations and extract them into accessible folders on the target systems.

Learn More:

- [Set Up Analysis Target](#)

Remote Collection on a FreeBSD*, QNX*, or Android* Target

For the FreeBSD, QNX, or Android targets, you install collectors from the target packages available in the complete installation of VTune Profiler.

Learn More:

- [FreeBSD Targets](#)
- [QNX Targets](#)
- [Android Targets](#)

Remote Collection on a Linux Target

Linux target packages do not contain the binaries necessary for direct collection on the target. These target packages must be executed from the host.

In the VTune Profiler GUI, you can configure a remote SSH target for automatic collection, without the need to copy results manually. Once the SSH connection is established, the VTune Profiler GUI can deploy the collector package automatically.

Configure Analysis

WHERE

Remote Linux (SSH)

✖ Cannot find product on the target system at ubuntu@ec2-52-40-248-112.us-west-2.compute.amazonaws.com:/tmp/vtune_profiler_2024.0.0.626834/. Make sure VTune Profiler installation directory on the remote system option in the Analysis Target tab is set to the correct path. Alternatively, you may use the --target-install-dir option to specify the correct path from command line.

✖ Press Deploy button to install the target package.

✖ VTune cannot detect remote machine configuration.

Retry

SSH destination

ubuntu@ec2-52-40-248-112.us-west-2.compute.amazonaws.com

VTune Profiler installation directory on the remote system

/tmp/vtune_profiler_2024.0.0.626834

Temporary directory on the remote system

/tmp

Press to deploy VTune Profiler target package to the remote system.

Deploy

Learn More:

- [Linux Targets](#)

Install VTune Profiler Server

Set up Intel® VTune™ Profiler as a web server, using a lightweight deployment intended for personal use or a full-scale corporate deployment supporting multi-user environment.

Relevant Terminology

Before you install and use VTune Profiler Server, get familiar with these terms to ensure a smooth experience:

VTune Profiler Server	VTune Profiler started as a web server and serving a web site to access the VTune Profiler GUI from remote client machines using a web browser.
User	User of the VTune Profiler Server.
User client system	A machine that the User is logged to and used to access the VTune Profiler Server via a web browser.
Target system	A machine, local or remote, that is profiled with the VTune Profiler.
VTune Profiler Agent	A piece of VTune Profiler software that runs on a target system.

System Requirements

VTune Profiler Server System

- 64-bit Linux* or Windows* OS
- Same system requirements and supported operating system distributions as specified for VTune Profiler command line tool in the [Release Notes](#)

Client System

- Chrome, Firefox or Safari (recent versions)

VTune Profiler Server is tested with the latest versions of supported browsers at the time of each release.

Target System

- 32- or 64-bit Linux or Windows OS
- Same system requirements and supported operating system distributions as specified for VTune Profiler target systems in the [Release Notes](#)

NOTE

When you install VTune Profiler Server on a particular platform, you must use profiling targets of the same platform. You cannot use VTune Profiler Server to profile across platforms.

VTune Profiler Server Deployment

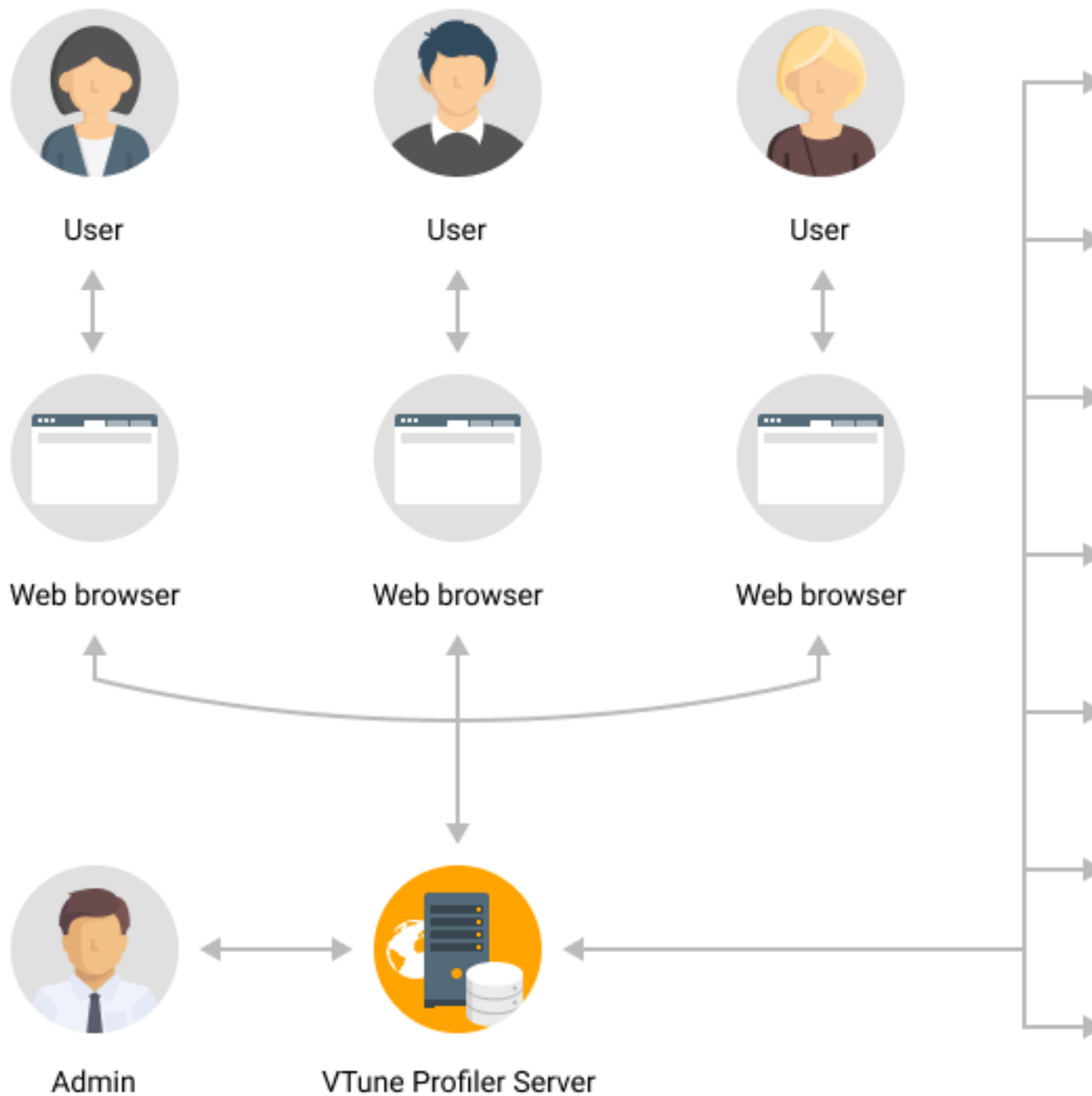
The deployment of VTune Profiler server depends on your usage mode and purpose:

Deployment Mode	Benefits	Limitations
Personal use/evaluation	• No host platform setup. VTune Profiler Server is installed as part of the VTune Profiler GUI. • Quick on-boarding experience with self-signed TLS certificates	• Single user mode • Medium security level
Integration with SAML Single-Sign-On (SSO)	• Automatic authentication with company accounts • Support for multi-user environment • High security level • Support for selective access to VTune Profiler Server (for example, per user network group)	• Mandatory company IT support to register VTune Profiler Server in IT SAML SSO infrastructure
Deployment behind a reverse proxy (NGINX*, Apache* web server, IIS, etc.)	• Reuse of existing IT web hosting infrastructure (including transport security and user authentication) • High security level • Support for multi-user environment	• DevOps expertise required

Depending on your choice, proceed with these steps:

- [Set up transport security.](#)
- [Configure user authentication/authorization.](#)

How It Works



1. (Reverse proxy and SAML SSO modes) Admin installs a VTune Profiler Server instance in a lab.

2. (Reverse proxy and SAML SSO modes) Admin emails the URL of the installed VTune Profiler Server to the User(s).
3. User accesses the VTune Profiler via a supported web browser, configures and runs analysis on an arbitrary target system.

VTune Profiler Server can be accessed from any client machine.

4. When analysis is initiated, the VTune Profiler Server installs a VTune Profiler Agent on the specified target system. This agent performs collection and uploads results to the VTune Profiler Server for analysis and storage.

Learn more about this service in [Web Server Interface](#).

Set Up Transport Security

VTune Profiler Server web site is accessible via encrypted HTTPS connection. HTTPS requires a Transport Layer Security (TLS) certificate. Depending on your [deployment mode](#), you can use different types of TLS certificates.

Self-Signed TLS Certificate

The self-signed certificate is automatically generated when the VTune Profiler Server is started. No additional actions are required from the user who starts the server, but the web browser will provide a warning that the server certificate is not trusted and will ask for a confirmation to proceed.

Signed TLS Certificate

You are recommended to use properly signed TLS certificates so that web browsers automatically validate authenticity of the VTune Profiler Server. Such certificate should be provisioned by your company IT department.

To set up the transport security, the Admin should follow these steps:

1. Provide the signed TLS certificate to users of the VTune Profiler Server.

Make sure to include the VTune Profiler Server DNS name to either **Common Name** or **Alternative Domain Names**.

For example, if the URL to access the VTune Profiler Server is `https://vtune.lab01.myorg.com`, the TLS certificate **Common Name** should be `vtune.lab01.myorg.com`, or `vtune.lab01.myorg.com` should be included into **Alternative Domain Names**.

2. Start the VTune Profiler Server as follows:

```
vtune-backend --tls-certificate /path/to/vtune.lab01.myorg.com.pfx --tls-certificate-password-path /path/to/cert_password.txt
```

You can also enter the certificate password interactively by using the `--tls-certificate-password` option instead of `--tls-certificate-password-path`. In this case, the VTune Profiler Server will prompt to enter the password:

```
vtune-backend --tls-certificate /path/to/vtune.lab01.myorg.com.pfx --tls-certificate-password
Certificate password:
```

If the certificate private key is stored in a separate file, use the `--tls-certificate-key` option:

```
vtune-backend --tls-certificate /path/to/vtune.lab01.myorg.com.crt --tls-certificate-key /path/to/vtune.lab01.myorg.com.key
```

See Also

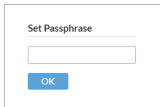
Configure User Authentication/Authorization

Use the default passphrase authentication to run the VTune Profiler Server, or benefit from your company solutions with reverse proxy or SAML authentication.

User authentication and authorization for VTune Profiler Server is controlled by a configuration file stored in `<vtune-install-dir>/backend/config.yml`. This configuration file uses YAML format and comes with brief inline documentation describing available configuration options.

Passphrase Authentication

In the default personal use mode, VTune Profiler Server is configured to use passphrase authentication/authorization. When you [start the server](#), you can specify a passphrase:



There are no usernames involved: if the passphrase is shared between multiple users, then they are treated as the same user.

VTune Profiler persists the hash of the passphrase. The browser also persists a secure HTTPS cookie so that you do not enter the passphrase each time. Cookie expiration time is configurable, default value is 365 days. When you access the VTune Profiler Server from a different machine or use a different browser, or if the browser cookies are cleaned / expired, then you are prompted to enter the passphrase again.

If you forget the passphrase, you can reset it by re-running the VTune Profiler Server using the `--reset-passphrase` option. The server provides an outcome URL with a one-time token to reset the passphrase:

```
vtune-backend --reset-passphrase
Serving GUI at https://127.0.0.1:65417?one-time-token=e2ed7c1365c972ec1024ac4e53179a08
```

When you open this URL in a web browser, you are prompted to set a new passphrase.

Reverse Proxy Authentication

VTune Profiler Server can be deployed behind a reverse proxy, which is a web server that forwards all requests to the VTune Profiler Server and serves its responses back to the user. With this type of setup, the system administrator can configure arbitrary user authentication and authorization in the reverse proxy. Reverse proxy is configured to pass authenticated user ID to the VTune Profiler Server, while the VTune Profiler Server is configured to trust this user ID.

To enable the reverse proxy authentication, the administrator needs to follow these steps:

1. Change the authentication type in the `<vtune-install-dir>/backend/config.yml` to `reverse-proxy` and specify the `header`, which is an HTTP header that reverser proxy uses to pass authenticated user ID.
2. Start the VTune Profiler Server as follows:
 - **If VTune Profiler Server and reverse proxy are on the same host:** start the VTune Profiler Server without the `--allow-remote-ui` option to prevent remote connections to be accepted by the VTune Profiler Agent:

```
vtune-backend --web-port=8080
Serving GUI at https://127.0.0.1:8080
warn: Server access is limited to localhost only. To enable remote access, restart with --allow-remote-ui.
```

- **If VTune Profiler Server and reverse proxy are on different hosts:** configure the reverse proxy to use a client certificate authentication when calling the VTune Profiler Server. Provide the VTune Profiler Server with the path to the public part of the reverse proxy client certificate :

```
vtune-backend --allow-remote-ui --client-certificate /path/to/public/reverse/proxy/cert.crt
```

NOTE

You are recommended to use the client certificate authentication even when VTune Profiler Server and the reverse proxy are on the same host to prevent an unauthorized access from the host system.

SAML SSO Authentication

VTune Profiler Server supports SAML 2.0 Single Sign On (SSO) for user authentication.

To enable the SAML SSO authentication, the Admin needs to follow these steps:

1. Change the authentication type in the `<vtune-install-dir>/backend/config.yml` to `saml` and specify the `rootUrl` and the `entityID`.
2. Request the IT service to register the VTune Profiler Server into the SAML SSO infrastructure. The request should include the entity ID, consume URL (`rootUrl + consumePath`), and the name of a network user group to be provided with an access to the VTune Profiler Server.

In response, the IT service provides the entry point for SAML Identity Provider and its public certificate.

3. Enter the data provided with the IT service to the `entryPoint` and `cert` fields in the `config.yml` file.
4. Start the VTune Profiler Server.

[See Also](#)

Understand Security Considerations

Performance profiling is an activity that may involve making important security decisions. Learn about some important security considerations before you use Intel® VTune™ Profiler.

Due to the inherent nature of performance profiling, Intel® VTune™ Profiler requires certain levels of access for proper operation of advanced analyses. Make sure you understand the implications of these accesses so that you can make informed security decisions.

Administrator and Root Privileges

VTune Profiler requires administrator or root privileges for performing specific types of analyses. On Windows* OS, this means starting VTune Profiler as Administrator, and on Linux* systems, this requires sudo privileges.

Only run VTune Profiler with elevated privileges if a specific analysis requires these privileges. Avoid staying in elevated mode for viewing collected results.

Controlling Sampling Driver Access (Linux* OS)

By default, on Linux OS, VTune Profiler installer creates a `vtune` user group, which is given access to the Sampling Driver through the Linux* I/O Control. It is recommended to not alter the default settings, for example, by creating a broad user group. Since the driver runs on the kernel level, exposing the driver to a large group of users can make your system vulnerable. Additionally, any user that has access to the driver can potentially obtain sensitive information by collecting performance metrics from the system.

Though VTune Profiler takes preemptive measures by validating all user input, it is recommended that you follow the principle of least required privilege when allowing access to the sampling driver.

Security Implications of Setting `perf_event_paranoid` (Linux* OS)

On Linux OS, the `perf_event_paranoid` setting controls the access levels for unprivileged users of `perf`. VTune Profiler may recommend that you set this value to 0 to perform a specific analysis. At this level, the collected data includes per-process and system-wide performance monitoring data, including CPU and system events both from the user space and the kernel. This may create a potential for sensitive data leaks.

For more information on the usage of `perf` with VTune Profiler and possible limitations, see a technical recipe on [Profiling Hardware Without Intel Sampling Drivers](#).

VTune Profiler Server Authentication Security

Though all network traffic of VTune Profiler Server is encrypted, it is important to select the appropriate authentication scheme when installing VTune Profiler Server. While passphrase authentication is a viable option for some use cases, such as personal use, it is recommended to use other authentication schemes offered when using VTune Profiler Server in broader environments. You can find more information in [Install VTune Profiler Server](#).

Update Your Intel® VTune™ Profiler

Use the Intel installer to check for newer versions of Intel® VTune™ Profiler and update when needed.

Windows OS

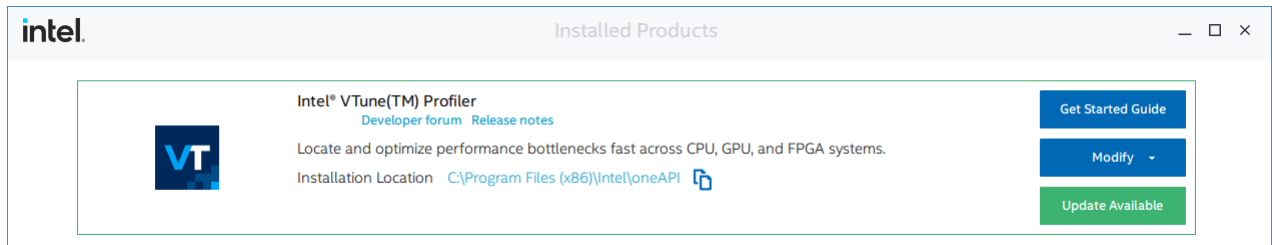
This procedure uses the Intel installer to check for product updates, for which you must have a stable connection to the internet.

If your connection is unreliable, consider [downloading an offline installer](#) that contains a newer version of VTune Profiler. Then follow the steps in [Install on Windows* OS](#) to complete your update.

To use the Intel installer:

1. On your Windows system, go to **Settings > Apps**.
2. In the list of installed applications, find VTune Profiler and click **Modify**.

The Intel product installer opens and checks if updates are available for all installed Intel software products.



3. If a newer version of VTune Profiler is available, click the **Update Available** button to review the newer version.
4. Click **Install** to upgrade your application.

Linux OS

Use the Installer

To update VTune Profiler using the installer, do one of the following:

- Reuse the installer left from a previous version.
- Download the installer from the [Download page](#).

Once you have the installer,

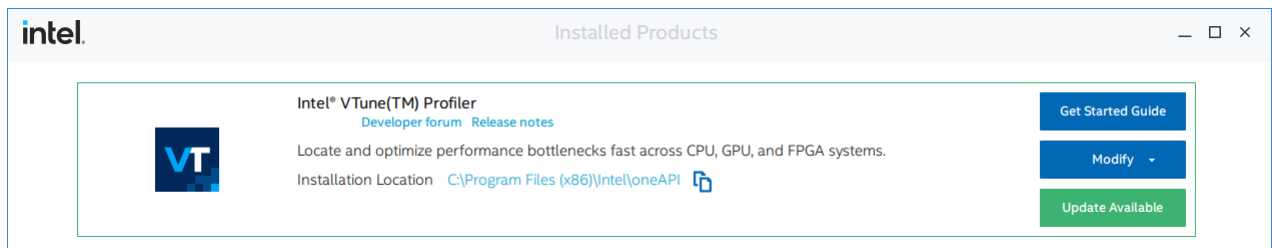
1. Open a terminal window as an administrator.
2. Go to the directory where you downloaded the installer.
3. If necessary, enable the execution of the installer. Run this command:

```
chmod +x <installer-package-name>.sh
```

4. Run the installer executable as a superuser:

```
sudo ./<installer-package-name>.sh
```

The installer window opens and checks for updates for all installed Intel® software products.



5. If a newer version of VTune Profiler is available, click the **Update Available** button to review the newer version.
6. Click **Install** to upgrade your application.

Use Package Managers

If you installed VTune Profiler using package managers, regular system-wide package updates keep VTune Profiler updated to the latest version without requiring any additional action.

To update VTune Profiler independently, use these commands:

APT:

```
sudo apt install intel-oneapi-vtune
```

YUM:

```
yum upgrade intel-oneapi-vtune
```

DNF:

```
dnf upgrade intel-oneapi-vtune
```

Zypper:

```
zypper up intel-oneapi-vtune
```

Frequently Asked Questions

How do I install the Sampling Drivers?

When you install Intel® VTune™ Profiler, the installation process should also install the required sampling drivers you require for data collection, as long as you complete the process with appropriate permissions (administrative/root/sudo access). Drivers are also installed on a target system when VTune Profiler connects to the target system during analysis configuration (administrative/root/sudo access required via a password-less SSH connection). If the drivers fail to install, you can configure them manually.

Learn More:

- [Sampling Drivers](#)
- [Install the Sampling Drivers for Windows* Targets](#)
- [Install the Sampling Drivers for Linux* Targets](#)
- [Install the Sampling Drivers for Android* Targets](#)

NOTE

Occasionally, updates to supported operating systems can cause the [sampling drivers](#) to fail during building or loading. In this case, before you update Intel® VTune™ Profiler, you may need to [download the latest versions](#) of the sampling drivers and manually [build the drivers](#).

What features are available for a non-root or non-administrator user?

Windows* OS

On a Windows machine, when you install and launch VTune Profiler as a regular user, the analysis types that employ User-Mode Sampling are available:

- [Hotspots analysis \(without microarchitecture performance insights\)](#)
- [Threading analysis](#)
- [Memory Consumption analysis](#)

Linux* OS

On a Linux system, VTune Profiler can use the capabilities of the Linux Perf* utility to deliver several features, without requiring sampling drivers and/or root privileges. Certain considerations, prerequisites, and configuration steps apply to the use of Linux Perf.

Learn More:

- A technical recipe on [Profiling Hardware Without Intel Sampling Drivers](#)